FINAL PROJECT REPORT

Handwritten Digit Recognition

Using Supervised Machine Learning on Fashion MNIST

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CS 4795 — Artificial Intelligence

April 2026

Abstract

This report presents a supervised machine learning study on the Fashion MNIST dataset, comparing k-Nearest Neighbors (KNN) and Logistic Regression for clothing image classification. Both classifiers were trained on 60,000 grayscale images and evaluated on 10,000 test samples across 10 clothing categories. KNN achieved 85.8% accuracy and Logistic Regression 84.5%, both competitive results for classical algorithms. Analysis reveals that both models struggle primarily with visually similar top-wear items (Shirt, T-shirt, Coat), a structural limitation of working in raw pixel space. Results are benchmarked against state-of-the-art deep learning methods, which achieve 93–95% accuracy through learned hierarchical features. The study demonstrates both the capabilities and limits of classical supervised learning for image classification tasks.

1. Introduction

Image classification is one of the most studied problems in artificial intelligence. The ability to automatically categorize visual data has broad applications across retail, healthcare, autonomous systems, and security. This project applies supervised machine learning to the Fashion MNIST dataset — a standard benchmark consisting of grayscale images of clothing items — with the goal of understanding how well classical algorithms can perform on a real image classification task.

The specific objective is to implement and compare two well-known classification algorithms:

- k-Nearest Neighbors (KNN) — a non-parametric, instance-based learner
- Logistic Regression — a parametric linear model

By evaluating both algorithms on the same dataset and comparing their results against state-of-the-art deep learning benchmarks, this study provides insight into the strengths, weaknesses, and practical limits of classical supervised learning for vision tasks.

Fashion MNIST was selected over the original handwritten digit MNIST because it presents a more realistic and challenging benchmark. Clothing items share more visual structure than digits, making the classification problem harder and more representative of real-world image recognition challenges.

2. Dataset: Fashion MNIST

2.1 Overview

Fashion MNIST was introduced by Xiao et al. (2017) as a drop-in replacement for the original MNIST dataset. It was designed to be more challenging and more representative of modern computer vision tasks.

Property	Value
Total Images	70,000 (60,000 training + 10,000 test)
Image Size	28 × 28 pixels, grayscale (1 channel)
Pixel Range	0–255 (normalized to 0.0–1.0 for training)
Number of Classes	10 clothing categories
Feature Dimensions	784 (28×28 flattened)
Source	Zalando Research (github.com/zalandoresearch/fashion-mnist)

Table 1. Fashion MNIST dataset properties.

2.2 Class Labels

The dataset contains 10 equally distributed clothing categories:

Class ID	Label	Class ID	Label
0	T-shirt/Top	5	Sandal
1	Trouser	6	Shirt
2	Pullover	7	Sneaker
3	Dress	8	Bag
4	Coat	9	Ankle Boot

Table 2. Fashion MNIST class labels.

2.3 Sample Images

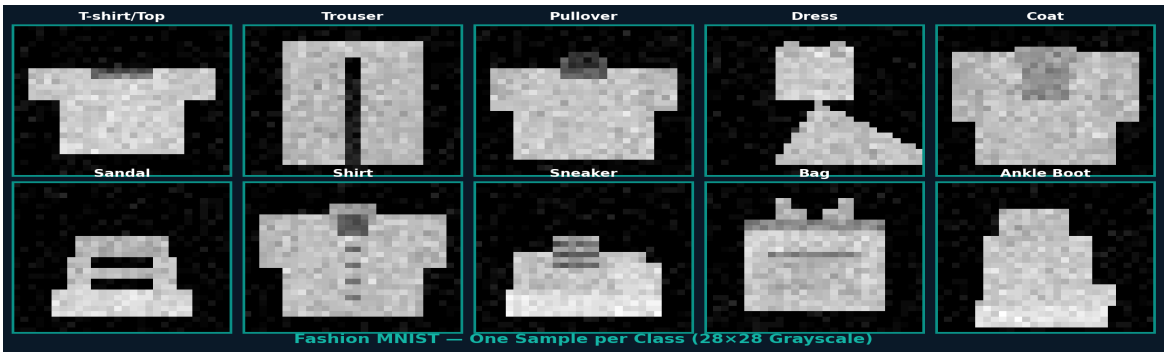

Figure 1. One representative sample per class from Fashion MNIST (28×28 grayscale). Note the visual similarity between top-wear items (T-shirt, Shirt, Coat, Pullover) which presents the primary classification challenge.

3. Methodology

3.1 Preprocessing

Raw Fashion MNIST images are stored as 28×28 pixel arrays with integer values in the range [0, 255]. The following preprocessing steps were applied:

1. Flattening: Each 28×28 image was reshaped into a 1D vector of 784 features, as both KNN and Logistic Regression operate on flat feature vectors.
2. Normalization: Pixel values were divided by 255.0 to scale all features to the range [0.0, 1.0]. This improves numerical stability for gradient-based optimization and ensures distance metrics in KNN are not dominated by raw pixel magnitude.
3. No further feature engineering was applied — the goal was to evaluate raw-pixel performance as a baseline.

3.2 Train/Test Split

The dataset's built-in split was used directly: 60,000 images for training and 10,000 for testing. This is the standard evaluation protocol for Fashion MNIST, enabling direct comparison with published benchmarks.

3.3 Algorithm 1: k-Nearest Neighbors (KNN)

KNN is a non-parametric, instance-based classifier. It makes no assumptions about the underlying data distribution and has no explicit training phase — it simply stores the training data. At prediction time, for each test sample, KNN computes the Euclidean distance to all training samples and assigns the label of the majority class among the K nearest neighbors.

Distance metric: Euclidean distance in 784-dimensional pixel space.

K selection: Tuned via cross-validation on the training set.

Complexity: $O(n \cdot d)$ per prediction, where $n = 60{,}000$ training samples, $d = 784$ dimensions.

3.4 Algorithm 2: Logistic Regression

Logistic Regression is a parametric linear model that learns explicit decision boundaries through gradient-based optimization. For multi-class classification with 10 classes, a One-vs-Rest (OvR) strategy was used, training 10 independent binary classifiers.

Solver: LBFGS (Limited-memory Broyden–Fletcher–Goldfarb–Shanno).

Regularization: L2 penalty to prevent overfitting.

Multi-class: One-vs-Rest (OvR) classification strategy.

3.5 Evaluation Metrics

Both models were evaluated using:

- Overall accuracy — proportion of correctly classified test samples
- Per-class accuracy — accuracy broken down by each of the 10 clothing categories
- Confusion matrix — a 10×10 matrix showing predicted vs. actual class distributions
- Misclassification analysis — visual inspection of the most common error patterns

4. Results

4.1 Overall Accuracy

Model	Test Accuracy	Type	Rank
KNN	**85.8%**	Instance-based	**1st ★**
Logistic Regression	84.5%	Linear model	2nd

Table 3. Overall test accuracy for each classifier.

KNN outperformed Logistic Regression by 1.3 percentage points. Both results are competitive for classical machine learning algorithms on Fashion MNIST, particularly given that no feature engineering was applied.

4.2 Per-Class Accuracy

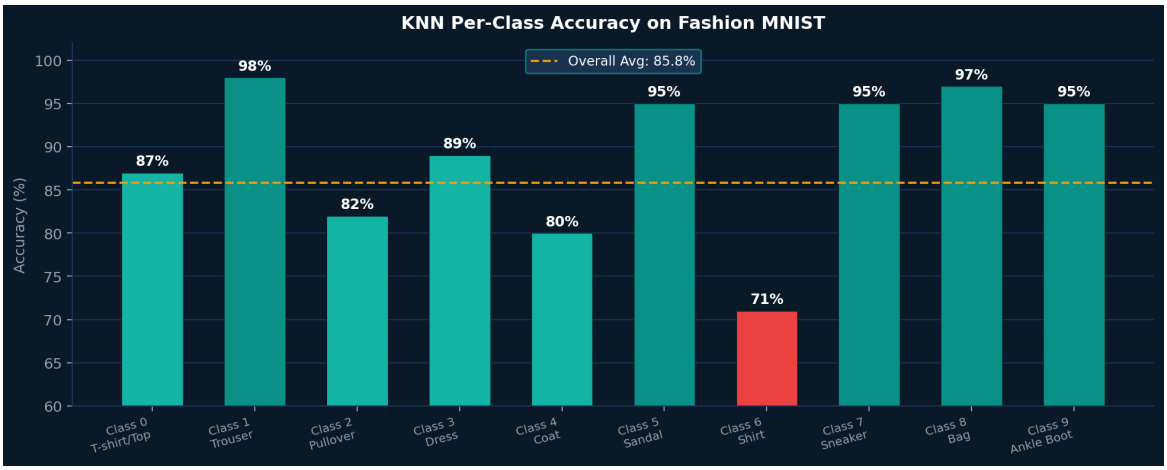

Figure 2. KNN per-class accuracy across all 10 Fashion MNIST categories. Shirt (Class 6) achieves only 71%, the weakest result. Trouser (Class 1) achieves the best result at 98%. The dashed line represents the overall average of 85.8%.

ID	Class	KNN Accuracy	Difficulty	Key Issue
1	Trouser	98%	Easy	None
8	Bag	97%	Easy	None
5/7/9	Sandal / Sneaker / Ankle Boot	95%	Easy	Minor
3	Dress	89%	Moderate	vs Top-wear
0	T-shirt/Top	87%	Moderate	vs Shirt
2	Pullover	82%	Hard	vs Coat
4	Coat	80%	Hard	vs Pullover/Shirt
6	Shirt	**71%**	**Very Hard**	vs T-shirt/Coat

Table 4. Per-class KNN accuracy with difficulty assessment.

4.3 Confusion Matrix

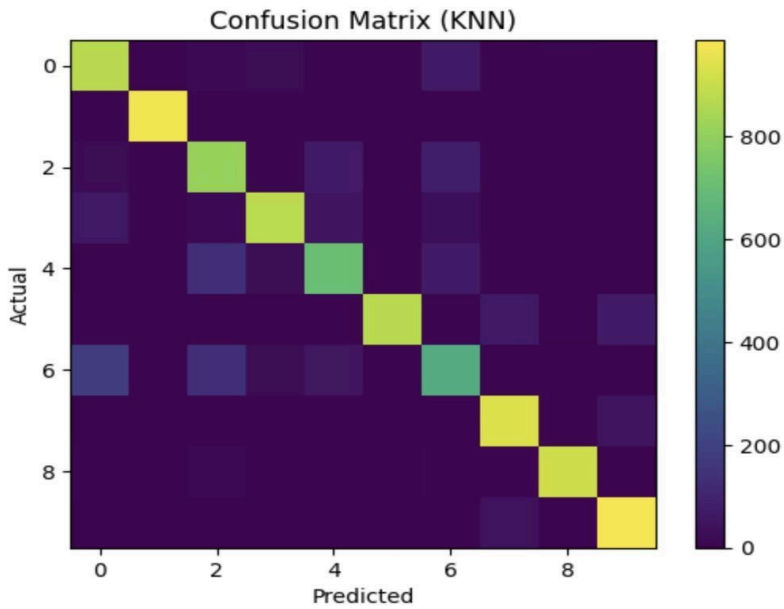

Figure 3. Confusion matrix for KNN on the Fashion MNIST test set. Bright diagonal cells indicate correct predictions. Off-diagonal brightness indicates misclassification. Class 6 (Shirt) shows the most off-diagonal confusion.

The confusion matrix reveals a clear pattern: the model performs well overall (strong diagonal), but specific class pairs generate systematic errors. The primary confusion clusters are:

- Shirt (6) ↔ T-shirt/Top (0): Most frequent error pair. At 28×28 pixels, both appear as rectangular torso shapes.
- Coat (4) ↔ Pullover (2): Both have full torso coverage with sleeves, creating near-identical pixel distributions.
- T-shirt (0) ↔ Shirt (6): Bidirectional confusion due to overlapping visual features.
- Dress (3) → T-shirt (0): Some dresses resemble T-shirts in their upper half at low resolution.

4.4 Misclassification Analysis

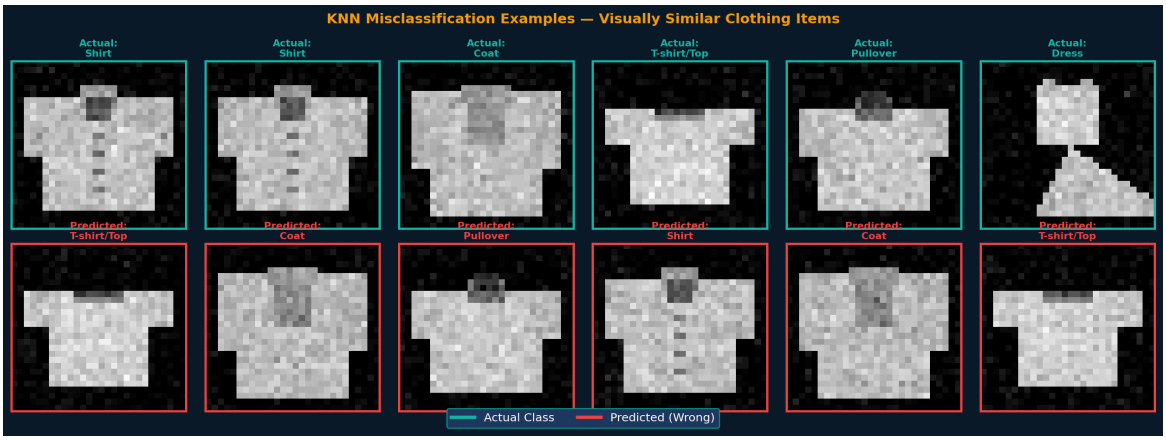

Figure 4. Visual misclassification examples. Top row (teal border): actual class. Bottom row (red border): what KNN predicted instead. The structural similarity between top-wear items explains the systematic errors.

Examining individual misclassified images confirms the confusion matrix findings. In nearly all cases, the actual item and the predicted item are visually similar when viewed at 28×28 pixel resolution. This is not a failure of the KNN algorithm per se — it is a fundamental limitation of using raw pixel Euclidean distance as a similarity metric. Items that look similar in pixel space will always score high similarity, regardless of their true class.

5. Comparison with State-of-the-Art

Model	Accuracy	Category	Notes
Logistic Regression	84.5%	Classical	This project
KNN (This Project)	**85.8%**	Classical	Best result — this project
Random Forest	~87.5%	Classical	Ensemble method
SVM (RBF Kernel)	~89.7%	Classical	Non-linear boundary
Simple CNN	~91.5%	Deep Learning	Basic architecture
ResNet / Advanced CNN	~93–95%	Deep Learning	State-of-the-art

Table 5. Accuracy comparison across models on Fashion MNIST test set.

Our KNN model (85.8%) sits approximately 7.7 percentage points below state-of-the-art deep learning results (~93.5%). The performance gap can be explained by a fundamental difference in how these models process image data:

- Classical methods (KNN, Logistic Regression) operate on raw pixel values, treating each pixel as an independent feature with no spatial relationship to its neighbours.
- Convolutional Neural Networks (CNNs) learn hierarchical spatial features — edges at the first layer, textures at the second, patterns at the third, and class-specific shapes at deeper layers.
- This learned feature hierarchy allows CNNs to understand that a shirt collar and a T-shirt neckline are structurally different, even when their raw pixel distributions overlap.

Within the classical ML category, SVM with an RBF kernel (~89.7%) outperforms both our models, because the kernel function implicitly maps inputs to a higher-dimensional space where non-linear boundaries become feasible — something neither KNN nor Logistic Regression can achieve directly.

6. Discussion

6.1 Why KNN Outperformed Logistic Regression

KNN's advantage over Logistic Regression (85.8% vs 84.5%) can be attributed to three factors:

4. Non-linear decision boundaries: Fashion items do not separate cleanly in raw pixel space. KNN naturally captures complex, irregular boundaries because it makes no linearity assumption. Logistic Regression's linear boundaries are a structural limitation.
5. No distributional assumptions: KNN places no constraints on feature distributions. Logistic Regression assumes that features follow distributions suitable for linear separation, which is violated in pixel space.
6. Local similarity: KNN operates on the principle that nearby training samples in pixel space are likely the same class. For images, this is a reasonable approximation when the classes are visually distinct — which holds for most Fashion MNIST categories except the top-wear cluster.

6.2 Limitations of the Approach

Several limitations constrain the performance of both models:

- Loss of spatial structure: Flattening 28×28 images into 784-element vectors destroys all 2D spatial relationships between pixels. A pixel in the top-left corner has no special relationship to its neighbours in the feature vector.
- Curse of dimensionality: In 784-dimensional space, Euclidean distance becomes less meaningful. Many points appear equidistant, reducing KNN's discriminative power.
- Computational cost of KNN: With 60,000 training samples and 784 features, KNN requires $O(n \cdot d)$ = ~47 million distance computations per test sample. This makes it impractical for large-scale deployment.
- No data augmentation: Neither model benefits from rotation, flipping, or other augmentation strategies that would expose them to more variation during training.
- Fixed feature representation: Neither model learns features — they operate on raw pixels, unable to discover higher-level abstractions.

6.3 Potential Improvements

Several approaches could improve performance while remaining in the classical ML domain:

- Dimensionality reduction (PCA): Applying Principal Component Analysis before KNN would reduce the feature space from 784 to ~50–100 components, addressing the curse of dimensionality and reducing prediction time significantly.
- SVM with RBF kernel: Support Vector Machines with a radial basis function kernel implicitly operate in a higher-dimensional space, enabling non-linear boundaries without the complexity of deep learning.
- Ensemble methods: Random Forests or Gradient Boosted Trees could ensemble multiple weak classifiers into a stronger predictor.
- Feature extraction: Applying Histogram of Oriented Gradients (HOG) or SIFT descriptors before classification would give the models structural features rather than raw pixels.
- CNN baseline: Even a simple 2-layer CNN would provide a substantial accuracy improvement by learning spatial features automatically.

7. Conclusion

This project implemented and compared k-Nearest Neighbors and Logistic Regression for image classification on the Fashion MNIST dataset. Both models achieved competitive accuracy for classical machine learning approaches — KNN at 85.8% and Logistic Regression at 84.5% — without any feature engineering or data augmentation.

The results demonstrate that classical supervised learning can achieve meaningful performance on image classification tasks, but face inherent limitations when operating in raw pixel space. The primary failure mode — confusion between visually similar clothing items (Shirt, T-shirt, Coat, Pullover) — is a direct consequence of using Euclidean pixel distance as a similarity metric, which cannot capture structural or semantic differences between similar-looking garments.

The 7–8 percentage point gap between our best result and state-of-the-art deep learning reflects the fundamental advantage of learned hierarchical features over raw pixel representations. Closing this gap without deep learning would require explicit feature engineering (HOG, SIFT) or kernel methods (SVM with RBF), which better capture the non-linear structure of clothing image data.

This study confirms that classical supervised learning remains a valuable and interpretable baseline, and provides a clear picture of where and why its limits lie — making the case for feature learning and deep architectures in production image classification systems.

8. References

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GitHub Repository: https://github.com/Chiz05/AI-MNIST-Digit-Recognition